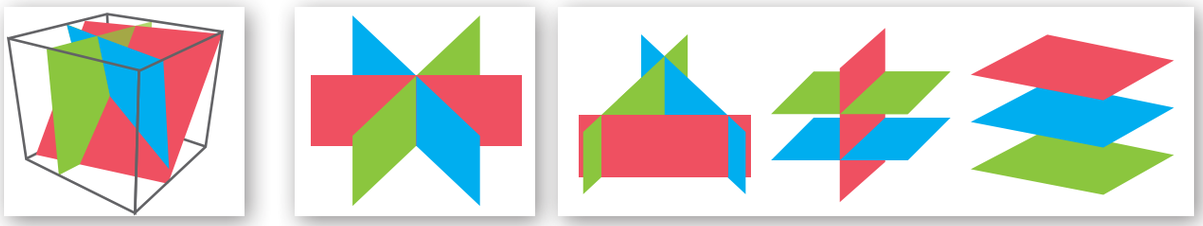
General Form: A system of linear equations in three variables $x,\ y,\ z$ has the general form

$$\begin{aligned} a_1x + b_1y + c_1z + d_1 &= 0 \\ a_2x + b_2y + c_2z + d_2 &= 0 \\ a_3x + b_3y + c_3z + d_3 &= 0 \end{aligned}$$

Each equation in the system represents a plane in three dimensional space and solution of the system of equations is precisely the point of intersection of the three planes defined by the three linear equations of the system. The system may have only one solution, infinitely many solutions or no solution depending on how the planes intersect one another.

The figures presented below illustrate each of these possibilities



only one solution | Infinitely many solutions | No solution

Fig. 3.4

Procedure for solving system of linear equations in three variables

Step 1: By taking any two equations from the given three, first multiply by some suitable non-zero constant to make the co-efficient of one variable (either x or y or z) numerically equal.

Step 2: Eliminate one of the variables whose co-efficients are numerically equal from the equations.

Step 3: Eliminate the same variable from another pair.

Step 4: Now we have two equations in two variables.

Step 5: Solve them using any method studied in earlier classes.

Step 6: The remaining variable is then found by substituting in any one of the given equations.

Note

- If you obtain a false equation such as $0=1$, in any of the steps then the system has no solution.
- If you do not obtain a false solution, but obtain an identity, such as $0=0$ then the system has infinitely many solutions.

Example 3.3 Solve the following system of linear equations in three variables $3x - 2y + z = 2$, $2x + 3y - z = 5$, $x + y + z = 6$.

Solution $3x - 2y + z = 2$ ...(1) $2x + 3y - z = 5$ ...(2) $x + y + z = 6$ ...(3)

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